ELECTRICAL POWER BUDGET & CONVERTER SIZING CALCULATION

Rigorous Numerical Proofs & Component Stress Sizing

1. Comprehensive System Load Budget

Subsystem Load	Supply Rail	Voltage	Avg Current	Peak Current	Peak Power	Operating Condition
ESP32-S3 DevKitC-1	5V_SYS	5.0V	160 mA	500 mA	2.50 W	Active WiFi/BLE peak current
MicroSD Breakout	5V_SYS / 3V3	3.3V/5V	30 mA	100 mA	0.50 W	Flash sector erase/write current
5V Buzzer & Status LED	5V_SYS	5.0V	15 mA	40 mA	0.20 W	Audible crash notification
NEO-M8N GNSS Breakout	3V3_SYS	3.3V	35 mA	65 mA	0.21 W	Continuous satellite acquisition
0.96-inch I2C OLED	3V3_SYS	3.3V	15 mA	30 mA	0.10 W	All white pixels illuminated
MPU6050 IMU Breakout	3V3_SYS	3.3V	3.8 mA	5.0 mA	0.02 W	DMP active 200Hz sampling
DS3231 Real-Time Clock	3V3_SYS	3.3V	0.15 mA	1.5 mA	0.005 W	TCXO temperature conversion
A7682S Cellular (Option A)	SIM_PWR	5.0V	250 mA	2000 mA	10.0 W	4G LTE Category 1 RF burst
TOTAL SYSTEM BUDGET	5V_SYS	5.0V	509 mA	2741 mA	13.54 W	Maximum concurrent load

2. Battery Current Draw Across Discharge Curve

The single 18650 Li-ion cell operates between 4.20V (100% SoC) and 3.00V (empty cutoff). Under maximum continuous system load (5V @ 0.70A = 3.50W) and maximum peak load (5V @ 2.74A = 13.70W with 90% boost efficiency eta):

Equation: I_BAT = P_OUT / (eta * V_BAT) = (V_OUT * I_OUT) / (eta * V_BAT)

Battery Voltage	Cont Power	Cont I_BAT	Peak Power	Peak I_BAT	Operating Assessment
4.20V (Full Charge)	3.50 W	0.925 A	13.70 W	3.62 A	Low thermal stress, maximum margin
3.70V (Nominal Plateau)	3.50 W	1.051 A	13.70 W	4.11 A	Standard cruise operating condition
3.30V (Low Battery Warn)	3.50 W	1.178 A	13.70 W	4.61 A	High-stress telemetry dispatch
3.00V (Emergency Cutoff)	3.50 W	1.296 A	13.70 W	5.04 A	Absolute worst-case peak current

[NOTE] BATTERY PROTECTION MOSFET VOLTAGE DROP VERIFICATION
Under worst-case 3.00V cutoff and 5.04A peak burst current, the dual paralleled FS8205A MOSFETs (RDSon = 12.5 mOhm combined) produce a total voltage drop of Delta_V = 5.04A * 0.0125 Ohm = 0.063V (63mV). The battery cell terminals remain at 3.00V - 0.063V = 2.937V, which is safely above the DW01A hard undervoltage cutoff threshold of 2.40V.

3. TPS61088 High-Current Boost Converter Sizing

1. Duty Cycle (D) at minimum input voltage (VIN = 3.0V, VOUT = 5.0V, eta = 0.90):

D = 1 - (VIN * eta / VOUT) = 1 - (3.0 * 0.90 / 5.0) = 0.460 (46.0%)

2. Average Inductor Current (IL_avg) at maximum continuous 5V/3.0A load:

IL_avg = (VOUT * IOUT) / (VIN * eta) = (5.0 * 3.0) / (3.0 * 0.90) = 5.56 A

3. Inductor Ripple Current (Delta_IL) with L = 1.0uH, Fsw = 600kHz:

Delta_IL = (VIN * D) / (Fsw * L) = (3.0V * 0.46) / (600,000 Hz * 1.0e-6 H) = 2.30 A

4. Peak Inductor Current (IL_peak):

IL_peak = IL_avg + (Delta_IL / 2) = 5.56A + 1.15A = 6.71 A

Selected Inductor: Bourns SRP6540-1R0M (Isat = 10.0A, Irms = 8.5A, DCR = 9.0 mOhm).

Saturation Safety Margin: (10.0A - 6.71A) / 10.0A = 32.9% margin under maximum continuous 3.0A load!

4. Cellular Transient Buffer Sizing (1000uF Polymer)

A7682S produces 2.0A current pulses for T_pulse = 577 microseconds during transmission bursts. The allowable voltage droop on the cellular rail is Delta_V_max = 150mV (to keep supply safely above the 3.4V reset threshold).

Capacitance required: C_req = (I_burst * T_pulse) / Delta_V_max = (2.0A * 577e-6 s) / 0.15V = 769 uF.

Selected: 1000uF / 6.3V Panasonic EEF-HX Ultra-Low ESR Polymer Capacitor (ESR = 12 mOhm).

ESR Voltage Droop: Delta_V_ESR = I_burst * ESR = 2.0A * 0.012 Ohm = 24.0 mV.

Capacitive Discharge Droop: Delta_V_cap = (2.0A * 577e-6) / 1000e-6 = 1.15 mV.

Total Transient Droop: Delta_V_total = 24.0mV + 1.15mV = 25.15 mV << 150 mV limit (83.2% safety margin).

5. Thermal Dissipation & Junction Temperatures

Active Component	Worst-Case Power Loss	Thermal Resistance	Temp Rise (Tamb=50°C)	Estimated Junction Temp
TPS2121 (Power Mux)	P = I^2 * R = (2.5)^2 * 0.056 = 0.35 W	Theta_JA = 48.6 °C/W	Delta_T = 17.0 °C	T_J = 67.0 °C (Max 125 °C)
BQ25895 (Charger NVDC)	P_loss = 3.0A * 0.15V = 0.45 W	Theta_JA = 35.0 °C/W	Delta_T = 15.8 °C	T_J = 65.8 °C (Max 125 °C)
TPS61088 (Boost 5V)	P_loss = 15W * (1 - 0.92) = 1.20 W	Theta_JA = 32.5 °C/W	Delta_T = 39.0 °C	T_J = 89.0 °C (Max 125 °C)
TPS62162 (Buck 3.3V)	P_loss = (3.3W / 0.90) - 3.3W = 0.37 W	Theta_JA = 65.0 °C/W	Delta_T = 24.1 °C	T_J = 74.1 °C (Max 125 °C)
Dual FS8205A (Protection)	P_loss = (5.04A)^2 * 0.0125 = 0.32 W	Theta_JA = 80.0 °C/W	Delta_T = 25.6 °C	T_J = 75.6 °C (Max 150 °C)